COURSE CODE/TITLE:	CPE-010: Data Structures and Algorithm	**SECTION:**	CPE21S3
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Seatwork 7.2: Algorithm Complexity

1. PROCEDURE

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Finding an Element
    - Best Case: O(1)
    - Worst Case: O(n)
    - Explanation: If the target is found at the first element requires one comparison then O1 is the best case. But if the target is at the last position then it requires all n elements to be checked making this the worst case.

2. Printing All Elements
    - Best Case: O(n)
    - Worst Case: O(n)
    - Explanation: Every element is so the loop always executes n times.

3. Nested Loops
    - Best Case: $O(n^2)$
    - Worst Case: $O(n^2)$
    - Explanation: Both loops will execute n times so it does it in $n \times n$.

4. Conditional Processing
    - Best Case: O(1)
    - Worst Case: O(n)
    - Explanation: When n is 100 or less then only the single output in the else statement executes making O(1) as the best case. When n is greater than 100 then the loop executes n times making O(n) as the worst case.

5. Fixed Inner Loop
    - Best Case: O(n)
    - Worst Case: O(n)
    - Explanation: The outer loop executes n times while the inner loop always executes 10 times. Since 10 is a constant, it simplifies to O(n) making it as both being the worst and best case.

6. Two Consecutive Loops
    - Best Case: O(n)
    - Worst Case: O(n)
    - Explanation: Both loops execute n times and it needs to print all outputs.

7. Dividing by Two
    - Best Case: O(log n)
    - Worst Case: O(log n)
    - Explanation: x is repeatedly divided by 2 until it reaches 1 so it requires $\log_2(n)$ repetitions making it both cases as O(log n).

8. Nested Loop With Different Limits
    - Best Case: $O(n^2)$
    - Worst Case: $O(n^2)$
    - Explanation: The inner loop depends on the value of the outer loop, making it the same as $O(n^2)$ as both cases.

9. Early Termination
    - Best Case: O(1)
    - Worst Case: O(n)
    - Explanation: The best case is if the target is equal to the first element making the loop stop. The worst case is if the target is at the last element.

10. Two-Level Search
    - Best Case: O(n)
    - Worst Case: $O(n^2)$
    - Explanation: O(n) will be the best case if the first element matches the target and the outer loop still runs n times. $O(n^2)$ as the worst case if the target is never found.

11. Linear Loop With Logarithmic Inner Work
    - Best Case: O(n log n)
    - Worst Case: O(n log n)
    - Explanation: The outer loop executes n times and during each iteration the inner loop performs O(log n) in both cases.

12. Conditional Nested Loop
    - Best Case: O(1)
    - Worst Case: $O(n^2)$
    - Explanation: If the first element equals the target then the nested loops are skipped and make O(1) as the best case. If the first element does not equal the target, both nested loops execute $O(n^2)$ times.

13. Finding the Maximum
    - Best Case: O(n)
    - Worst Case: O(n)
    - Explanation: If the first element is the maximum it will still check all elements making O(n) in both cases.

14. Three Nested Loops
    - Best Case: $O(n^3)$
    - Worst Case: $O(n^3)$
    - Explanation: All three loops always execute n times in both cases.

15. Comprehensive Analysis
    - Best Case: O(1)
    - Worst Case: O(n log n)
    - Explanation: In the best case then the target is found during the first iteration then ends the loop. O(n log n) as worst case if the outer loop executes n times, and the while loop takes O(log n) during each time.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Purpose of Use: _____

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